

Algorithm for Decomposition Using Functional Dependencies — DBMS

Definition: Decomposition Algorithms split a complex relational schema R into smaller sub-schemas ($R_1, R_2, \dots, R_n$) to eliminate anomalies while preserving **Lossless-Join** and **Functional Dependencies**.

1. Detailed Technical Explanation

1. Attribute Closure Algorithm (F^+)

The closure of an attribute set X under F , denoted X^+ , is the set of all attributes functionally determined by X .

Algorithm Steps:

```
Input: Set of attributes X, Set of FDs F
Output:  $X^+$  (Closure of X)

1. Set  $X^+ = X$ 
2. Repeat until no more attributes are added:
   For each FD  $(Y \rightarrow Z)$  in  $F$ :
     If  $Y \subseteq X^+$ :
        $X^+ = X^+ \cup Z$ 
3. Return  $X^+$ 
```

2. Lossless-Join Decomposition Testing

A decomposition of R into R_1 and R_2 is **lossless-join** with respect to F if and only if:

$$(R_1 \cap R_2) \rightarrow R_1 \quad \text{OR} \quad (R_1 \cap R_2) \rightarrow R_2$$

(Meaning: The common attributes between R_1 and R_2 must form a Super Key for at least one of the decomposed relations).

3. 3NF Synthesis Algorithm (Lossless-Join & Dependency Preserving)

```
Input: Relation R, Set of FDs F
Output: 3NF Decomposition of R

1. Compute Minimal Cover  $F_c$  for  $F$ .
2. For each FD  $(X \rightarrow Y)$  in  $F_c$ :
   Create a schema  $R_i = X \cup Y$ .
3. If no schema  $R_i$  contains a candidate key of  $R$ :
   Create an additional schema containing any candidate key of  $R$ .
4. Eliminate redundant schemas (if  $R_i \subseteq R_j$ , remove  $R_i$ ).
```

2. Core Concepts & Memory Keywords

- **Minimal Cover (Canonical Cover):** A simplified set of FDs with single attributes on the right-hand side and no extraneous attributes.
 - **Lossless Join:** Guarantees $R = R_1 \bowtie R_2$ (no spurious tuples created upon joining).
 - **Dependency Preservation:** Checks if $(F_1 \cup F_2 \cup \dots \cup F_n)^+ = F^+$.
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3. Must-Write Points for Exams

- Lossless join ensures that joining decomposed tables produces the exact original dataset without fake data.
 - Dependency preservation allows database engines to enforce constraints without performing expensive table joins.
 - 3NF synthesis algorithm guarantees both **lossless-join** and **dependency preservation** simultaneously.
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4. Quick Recall Flow

Find Minimal Cover F_c -> Compute Attribute Closures X^+ -> Test Lossless $(R_1 \bowtie R_2 \rightarrow R_1)$ -> Preserve FDs -> 3NF Synthesis